



An electronic tongue that can identify chemical makeup of any liquid

IBM research scientists have developed an electronic tongue with AI assistance that can identify the complicated chemical makeup of any liquid. The ‘tongue,’ known as



Hypertaste (Credit: IBM Research)

Hypertaste, is a reasonably tiny and affordable device. The final product employs an array of sixteen conductive polymeric sensors. To use Hypertaste, one has to submerge the array of sensors in the liquid. The sensors generate a series of voltage signals in response to the liquid’s compounds, which are referred to as the liquid’s chemical fingerprint. These are then transmitted to the user’s mobile device, where they are compared by a trained ML algorithm to a database of chemical compositions.

The new mechanism would help robots see in dark and carry out tasks

Rice University engineers have devised a mechanism for people to assist robots in ‘seeing’ their surroundings and

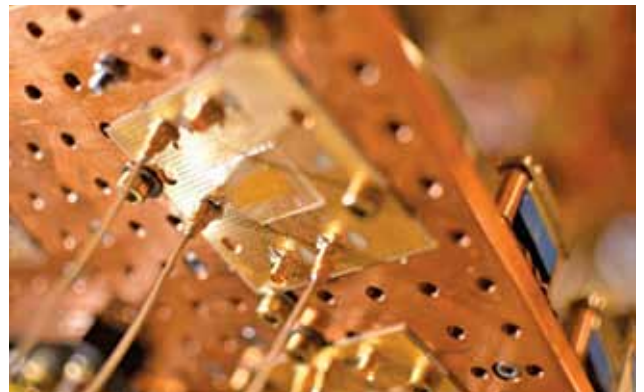


The task set for this Fetch robot by Rice University computer scientists is made easier by their BLIND software, which allows for human intervention when the robot’s path is blocked by an obstacle (Credit: Kavraki Lab)

carrying out tasks. This technique is called the Bayesian Learning IN the Dark (BLIND). The goal was achieved by integrating Bayesian inverse reinforcement learning with well-established motion planning approaches. When obstructions obscure a machine’s ‘view’ of its destination, BLIND inserts a human mid-process to modify the choreographic possibilities—or best guesses—offered by the robot’s algorithm, rather than setting a trajectory up front. This means that this technique simplifies human-robot relationships by incorporating human preferences.

Sound waves can now be controlled and modulated with a chip

For the first time ever, scientists from Harvard’s John A. Paulson School of Engineering and Applied Sciences (SEAS) have successfully controlled and modulated sonic



A chip that can control and modulate acoustic waves (Credit: Linbo Shao/Harvard SEAS)

waves using an electric field on chip. They showed that we can control acoustic waves on an integrated lithium niobate platform, bringing us one step closer to an acoustic integrated circuit. This chip is also called an on-chip electro-acoustic modulator. The modulator can alter the acoustic waves’ phase, amplitude, and frequency by imposing an electric field. The researchers are currently striving to construct more intricate, large-scale acoustic-wave circuits and links with more quantum systems.

This setup can calculate azimuth and elevation of sound source like an owl

An owl processes the difference in the arrival of a sound in each ear to calculate the azimuth and the elevation of the sound source. Scientists in France have incorporated these two aspects of computation into hardware that mim-